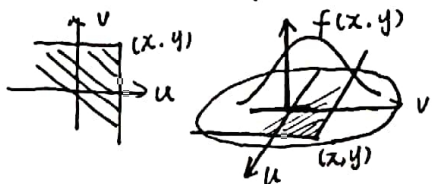


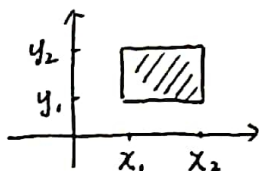
①

3.1.1 二维随机变量及其分布函数

 E 为随机变量. Ω 为样本空间. X, Y 是 Ω 的两个变量. (X, Y) 为二维随机向量(变量)分布函数: $F(x, y) = P\{X \leq x, Y \leq y\}$ X 与 Y 的联合分布函数性质 ① $0 \leq F(x, y) \leq 1$ ② $F(x, y)$ 是 x, y 的不减函数.③ $F(-\infty, y) = 0$ $F(x, -\infty) = 0$ $F(-\infty, -\infty) = 0$ $F(+\infty, +\infty) = 1$ ④ $F(x, y)$ 分别关于 x, y 分别右连续⑤ $x_1 < x_2, y_1 < y_2$

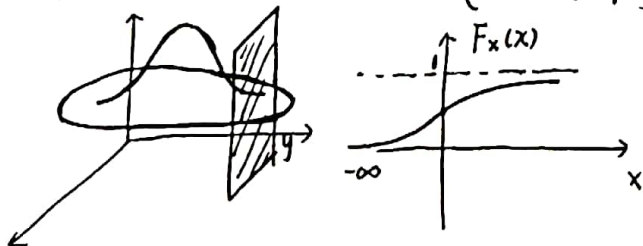
$$P\{x_1 < X \leq x_2, y_1 < Y \leq y_2\}$$

$$= F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1)$$



$$F_X(x) = P(X \leq x) = F(x, +\infty) = P(X \leq x, Y < +\infty)$$

$$F_Y(y) = P(Y \leq y) = F(+\infty, y) = P(X < +\infty, Y \leq y)$$



3.1.2 二维离散型的联合分布和边缘分布.

 X, Y 取离散值.

$X \backslash Y$	1	2	3	
1	0	$\frac{1}{2}$	$\frac{1}{8}$	→ 联合分布表. $P\{X=x_i, Y=y_j\}$ $= P_{ij}$
2	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	

(1) $P_{ij} \geq 0$ (2) $\sum_i \sum_j P_{ij} = 1$

$$F(x, y) = P\{X \leq x, Y \leq y\} = \sum_{X_i \leq x} \sum_{Y_j \leq y} P_{ij}$$

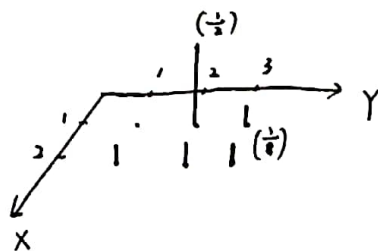
例) $F(1, -2) = P\{X \leq 1, Y \leq -2\} = 0$

$$F(1, 2) = P\{X \leq 1, Y \leq 2\} = \frac{1}{2}$$

$$F(4, 5) = 1$$

$$F(1.5, 2.6) = P\{X \leq 1.5, Y \leq 2.6\} = \frac{1}{2}$$

$X \backslash Y$	1	2	3
1	$\frac{1}{8}$	$\frac{1}{2}$	$\frac{1}{8}$
2	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$



边缘分布

$X \backslash Y$	1	2	3	$P_{i \cdot}^{(1)}$
1	0	$\frac{1}{2}$	$\frac{1}{8}$	→ $\frac{5}{8}$
2	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	→ $\frac{3}{8}$
$P_{\cdot j}^{(2)}$	$\frac{1}{8}$	$\frac{5}{8}$	$\frac{1}{4}$	△ 对行、列分别求和.



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X	1	2
P	$\frac{5}{8}$	$\frac{3}{8}$

Y	1	2	3
P	$\frac{1}{8}$	$\frac{5}{8}$	$\frac{1}{4}$

对行求和得 X 的边缘分布

对列求和得 Y 的边缘分布.

① 联合分布可唯一确定边缘分布.

② 边缘分布不能确定联合分布.

X, Y 独立时边缘分布能确定联合分布.

3.1.3 二维连续的联合密度和边缘密度.

$$F(x, y) = P(X \leq x, Y \leq y)$$

$$= \int_{-\infty}^x \int_{-\infty}^y f(s, t) ds dt$$

分布函数 $f(x, y)$ 联合密度分布函数.

性质 (1) $f(x, y) \geq 0$

$$(2) \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1$$

$$(3) \frac{\partial^2 F(x, y)}{\partial x \partial y} = f(x, y)$$

(4) G 为 XY 平面上的一个区域.

$$P\{(X, Y) \in G\} = \iint_G f(x, y) dx dy$$



例:

$$f(x, y) = \begin{cases} c & (x, y) \in G \\ 0 & \text{else} \end{cases}$$

$$G: x^2 + y^2 \leq r^2$$

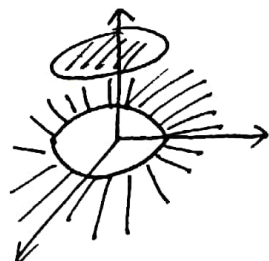
$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy$$

$$= \iint_G c dx dy$$

$$= c \cdot \pi r^2$$

$$\therefore c = \frac{1}{\pi r^2}$$

均匀分布.



$$\text{一维 } f(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{else} \end{cases} \quad \frac{1}{\text{长度}}$$

$$\text{二维 } f(x, y) = \begin{cases} \frac{1}{S(G)} & (x, y) \in G \\ 0 & \text{else} \end{cases} \quad \frac{1}{\text{面积}}$$

例:

$$f(x, y) = \begin{cases} e^{-(x+y)} & x > 0, y > 0 \\ 0 & \text{else} \end{cases}$$

(1) X, Y 的分布函数 $F(x, y)$

$$(2) \begin{matrix} \text{图} & x+y=1 \\ \text{图} & P\{(X, Y) \in G\} \end{matrix}$$

(3) $F_X(x) F_Y(y)$

$$(1) F(x, y) = P\{X \leq x, Y \leq y\}$$

$$\begin{aligned} x > 0, y > 0 \quad F(x, y) &= \int_{-\infty}^x \int_{-\infty}^y f(s, t) ds dt \\ &= \int_0^x \int_0^y e^{-s} e^{-t} ds dt \end{aligned}$$



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$$F(x, y) = \int_0^x e^{-s} ds \int_0^y e^{-t} dt$$

$$= (1 - e^{-x})(1 - e^{-y})$$

$$x, y < 0 \quad F(x, y) = 0$$

$$(2) P\{(x, y) \in G\} = \iint_G f(x, y) dx dy$$

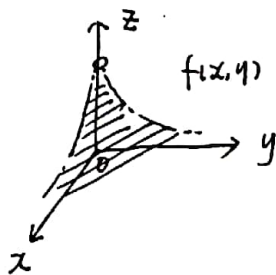
$$= \iint_G e^{-(x+y)} dx dy$$

$$= \int_0^1 dx \int_0^{1-x} e^{-(x+y)} dy$$

$$B) F_x(x) = \lim_{y \rightarrow +\infty} F(x, y)$$

$$= 1 - e^{-x} \quad x > 0$$

$$x \leq 0 \text{ 时 } F_x(x) = 0$$



边缘密度函数

$$F_x(x) = F(x, +\infty) = \int_{-\infty}^x \left[\int_{-\infty}^{+\infty} f(s, t) dt \right] ds$$

$$f_x(x) = \int_{-\infty}^{+\infty} f(x, t) dt = \int_{-\infty}^{+\infty} f(x, y) dy$$

$$f_y(y) = \int_{-\infty}^{+\infty} f(s, y) ds = \int_{-\infty}^{+\infty} f(x, y) dx$$

$$(3) f(x, y) = \frac{1}{\pi^2(1+x^2)(1+y^2)} \text{ 求 } f_x(x) f_y(y)$$

$$f_x(x) = \int_{-\infty}^{+\infty} \frac{1}{\pi^2(1+x^2)(1+y^2)} dy$$

$$= \frac{1}{\pi^2(1+x^2)} \arctan y \Big|_{-\infty}^{+\infty}$$

$$= \frac{1}{\pi(1+x^2)}$$

$$f_y(y) = \int_{-\infty}^{+\infty} \frac{dx}{\pi^2(1+x^2)(1+y^2)}$$

$$= \frac{1}{\pi(1+y^2)}$$

$$f(x, y) = f_x(x) \cdot f_y(y) \text{ 独立.}$$

$$(3) f(x, y) = \begin{cases} \frac{1}{\pi r^2} & x^2 + y^2 \leq r^2 \\ 0 & \text{else} \end{cases}$$

$$= f_x(x) \cdot f_y(y)$$

$$\text{证: 当 } |x| \leq r \text{ 时 } f_x(x) = \int_{-\infty}^{+\infty} f(x, y) dy$$

$$f_x(x) = \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} \frac{1}{\pi r^2} dy$$

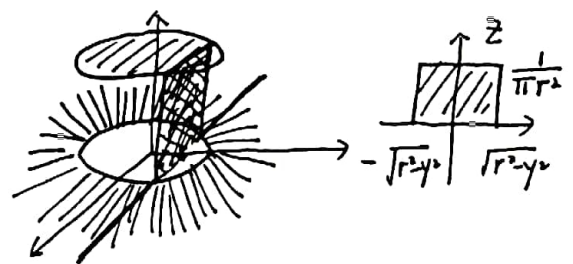
$$= \frac{2\sqrt{r^2-x^2}}{\pi r^2}$$

$$\text{当 } |x| > r \text{ 时}$$

$$f_x(x) = 0$$

$$f_x(x) = \begin{cases} \frac{2\sqrt{r^2-x^2}}{\pi r^2} & |x| \leq r \\ 0 & \text{else.} \end{cases}$$

$$f_y(y) = \begin{cases} \frac{2\sqrt{r^2-y^2}}{\pi r^2} & |y| \leq r \\ 0 & \text{else} \end{cases}$$



$$(3) f(x, y) = \begin{cases} 2(x+y-2xy) & 0 \leq x, y \leq 1 \\ 0 & \text{else} \end{cases}$$

$$\textcircled{1} 0 \leq x \leq 1 \quad f_x(x) = \int_{-\infty}^{+\infty} f(x, y) dy$$

$$= \int_0^1 2(x+y-2xy) dy$$

$$= 1$$



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$$\text{else } f_X(x) = 0$$

① 二维正态分布的边缘分布也是正态分布

② 两个边缘分布为正态分布，其二维随机变量并非一定为二维正态分布。

3.2.1 条件分布

$$F(x) = P(X \leq x)$$

$$F(x|A) = P\{X \leq x | A\}$$

例: $f(x) = \frac{1}{\pi(1+x^2)}$ 在 $x > 1$ 的条件下的条件分布

$$\text{解: } x > 1 \quad F(x|x > 1) = P\{X \leq x | X > 1\}$$

$$= \frac{P(X \leq x, X > 1)}{P(X > 1)}$$

$$x \leq 1 \text{ 时, } F(x|x > 1) = 0$$

$$x > 1 \text{ 时, } P\{1 < X \leq x\}$$

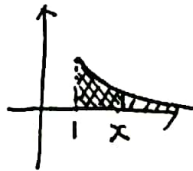
$$P\{1 < X \leq x\} = \int_1^x \frac{1}{\pi(1+t^2)} dt$$

$$= \frac{1}{\pi} \arctan t \Big|_1^x$$

$$= \frac{1}{\pi} \arctan x - \frac{1}{4}$$

$$P\{X > 1\} = \int_1^{+\infty} \frac{1}{\pi(1+t^2)} dt = \frac{1}{4}$$

$$F(x|x > 1) = \begin{cases} 0 & x \leq 1 \\ \frac{4}{\pi} \arctan x - \frac{1}{4} & x > 1 \end{cases}$$



3.2.2 离散型的条件分布

$X \backslash X_2$	0	1	$P_i^{(1)}$
0	0.1	0.3	0.4
1	0.3	0.3	0.6
$P_j^{(2)}$	0.4	0.6	

$$X_1=0 \quad P\{X_2=0 | X_1=0\}$$

$$= \frac{0.1}{0.4} = 0.25$$

$$X_1=0 \quad P\{X_2=1 | X_1=0\}$$

$$= \frac{0.3}{0.4} = 0.75$$

$$X_1=1 \quad P\{X_2=0 | X_1=1\} = \frac{0.3}{0.6} = 0.5$$

$$X_1=1 \quad P\{X_2=1 | X_1=1\} = \frac{0.3}{0.6} = 0.5$$

X_2	0	1
$P\{X_2 X_1=0\}$	0.25	0.75

X_2	0	1
$P\{X_2 X_1=1\}$	0.5	0.5

$$X_2=0 \quad P(X_1=0 | X_2=0) = \frac{0.1}{0.4} = 0.25$$

$$P(X_1=1 | X_2=0) = \frac{0.3}{0.4} = 0.75$$

$$P\{X=X_i | Y=y_j\} = \frac{P_{ij}}{P_j^{(2)}}$$

3.2.3 连续型的条件分布

定义: (X, Y) 是二维连续型随机变量。

$f(x, y)$ 为密度函数

边缘密度 $f_X(x)$ 、 $f_Y(y)$ 密度已知。

若 $f_Y(y) > 0$ 在 $Y=y$ 的条件下。

$$F(x|y) = \frac{\int_{-\infty}^x f(u, y) du}{f_Y(y)}$$

$$f(x|y) = \frac{f(x, y)}{f_Y(y)}$$

$$F(y|x) = \frac{\int_{-\infty}^y f(x, v) dv}{f_X(x)}$$

$$f(y|x) = \frac{f(x, y)}{f_X(x)}$$



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$$13.1: f(x, y) = \frac{1}{\pi^2(1+x^2)(1+y^2)}$$

$$f_x(x) = \frac{1}{\pi(1+x^2)}$$

$$f_y(y) = \frac{1}{\pi(1+y^2)}$$

$$f(x|y) = \frac{f(x, y)}{f_y(y)} = \frac{1}{\pi(1+x^2)}$$

$$f(y|x) = \frac{f(x, y)}{f_x(x)} = \frac{1}{\pi(1+y^2)}$$

$$13.1 \quad f(x, y) = \begin{cases} \frac{1}{\pi r^2} & x^2 + y^2 \leq r^2 \\ 0 & \text{else} \end{cases}$$

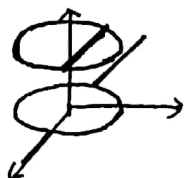
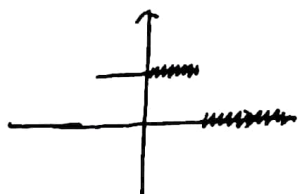
$$f_x(x) = \begin{cases} \frac{2\sqrt{r^2-x^2}}{\pi r^2} & |x| \leq r \\ 0 & \text{else} \end{cases}$$

$$f_y(y) = \begin{cases} \frac{2\sqrt{r^2-y^2}}{\pi r^2} & |y| \leq r \\ 0 & \text{else} \end{cases}$$

$$|y| < r \quad f(x|y) = \frac{f(x, y)}{f_y(y)}$$

$$= \begin{cases} \frac{1}{2\sqrt{r^2-y^2}} & -\sqrt{r^2-y^2} \leq x \leq \sqrt{r^2-y^2} \\ 0 & \text{else} \end{cases}$$

$$P(X > 0 | Y = 0) = \frac{1}{2}$$



$$P(X \leq x | Y = y)$$

$$= \frac{P(X \leq x, Y = y)}{P(Y = y)}$$

$$= \lim_{\epsilon \rightarrow 0} \frac{P(X \leq x, y \leq Y \leq y + \epsilon)}{P(y \leq Y \leq y + \epsilon)}$$

$$= \lim_{\epsilon \rightarrow 0} \frac{\int_{-\infty}^x \int_y^{y+\epsilon} f(u, v) du dv \cdot \frac{1}{\epsilon}}{\int_y^{y+\epsilon} f_y(v) dv \cdot \frac{1}{\epsilon}}$$

积分
中值定理

$$\int_y^{y+\epsilon} f_y(v) dv = f_y(\xi) \cdot \epsilon$$

$\xi \in (y, y + \epsilon)$ 且 $\epsilon \rightarrow 0$

$$\text{原式} = \lim_{\epsilon \rightarrow 0} \frac{\int_{-\infty}^x f(u, v) du}{f_y(y)}$$

3.2.4 随机变量的独立性

$$f(x|y) = f_x(x) = \frac{f(x, y)}{f_y(y)}$$

$$f(x, y) = f_x(x) \cdot f_y(y)$$

$$F(x, y) = F_x(x) F_y(y)$$

独立的充要条件为

$$P\{X \in S_X, Y \in S_Y\} = P\{X \in S_X\} \cdot P\{Y \in S_Y\}$$

1) = 二维离散型随机变量的独立性

$$P\{X = x_i, Y = y_j\} = P\{X = x_i\} P\{Y = y_j\}$$

X \ Y	0	1
0	0.2	0.4
1	0.2	0.6

$$0.4 \times 0.4 \neq 0.2$$

不独立



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$X \setminus Y$	0	1	
0	0.2	0.3	0.5
1	0.2	0.3	0.5
	0.4	0.6	

相乘为交点对应值

故独立

2) 二维连续型 $f(x, y) = f_x(x) f_y(y)$

例: 经理 8-12时 独立的.

秘书 7-9时

求时间相差不超过5分钟的概率

解: $X \quad f_X(x) = \begin{cases} \frac{1}{4} & 8 < x < 12 \\ 0 & \text{else} \end{cases}$

$Y \quad f_Y(y) = \begin{cases} \frac{1}{2} & 7 < y < 9 \\ 0 & \text{else} \end{cases}$

X, Y 独立.

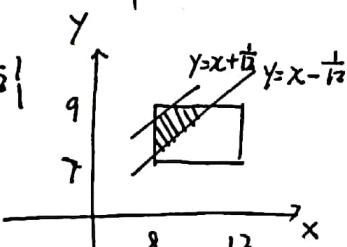
$f(x, y) = f_X(x) f_Y(y) = \begin{cases} \frac{1}{8} & 8 < x < 12 \\ & 7 < y < 9 \\ 0 & \text{else} \end{cases}$

$P(|X - Y| \leq \frac{1}{12})$

$= \iint_G f(x, y) dx dy$

$= \frac{1}{8} \iint_G 1 dx dy$

$= \frac{1}{8} \times \frac{1}{6} = \frac{1}{48}$



若变量独立, 由变量构造的函数也独立.

定理: X, Y 独立, $g_1(X), g_2(Y)$ 也是独立的.

X, Y 独立, 则 X^2, Y^2 独立. $a_1 X + b_1, a_2 X + b_2$ 独立.

$a_3 Y + b_3$ 独立.

3.3 二维随机变量函数的分布.

(1) 二维离散型.

$X \setminus Y$	4	4.2
5	0.2	0.4
5.1	0.3	0.1

土地长宽为 X, Y . * $Z = XY$

Z	20	21	20.4	21.42
P	0.2	0.4	0.3	0.1

例: X_1, X_2 独立. 0-1分布. 参数为 P

求 $X_1 + X_2$

X_1	0	1
P	$1-P$	P

X_2	0	1
P	$1-P$	P

$X_1 + X_2$	0	1	2
P	$(1-P)^2$	$2P(1-P)$	P^2

▲ $X_1 + X_2 \sim B(2, P)$ 结论值得记忆.

例: X, Y 独立是 λ_1, λ_2 的泊松分布

求 $Z = \frac{X+Y}{X+Y}$. $P\{X=k\} = \frac{\lambda_1^k}{k!} e^{-\lambda_1}$

解: $\{Z=k\} = \sum_{i=0}^k \{X=i, Y=k-i\}$

$P\{Z=k\} = \sum_{i=0}^k P\{X=i, Y=k-i\}$

$= \sum_{i=0}^k P\{X=i\} P\{Y=k-i\}$

$= \sum_{i=0}^k \frac{\lambda_1^i}{i!} e^{-\lambda_1} \cdot \frac{\lambda_2^{k-i}}{(k-i)!} e^{-\lambda_2}$

$= \frac{(\lambda_1 + \lambda_2)^k}{k!} e^{-(\lambda_1 + \lambda_2)}$

▲ $\lambda_1 + \lambda_2$ 泊松分布的可加性.



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3.3.2 二维连续变量函数的分布

已知 (X, Y) , $f(x, y)$, 有 $Z = g(X, Y)$, 求 Z 的密度函数及分布.

$$\begin{aligned} 1) F_Z(z) &= P\{Z \leq z\} = P\{g(X, Y) \leq z\} \\ &= \iint_{D_z} f(x, y) dx dy \end{aligned}$$

$$D_z = \{(x, y) | g(x, y) \leq z\}$$

(2) 上式两侧同时求导

$$f_Z(z)$$

例: $f(x, y) = \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}} \quad Z = \sqrt{x^2+y^2}$

解: 1) $Z < 0$ 时

$$F_Z(z) = P\{Z \leq z\} = P\{\sqrt{x^2+y^2} \leq z\} = 0$$

2) $Z \geq 0$ 时

$$F_Z(z) = P\{Z \leq z\} = P\{\sqrt{x^2+y^2} \leq z\}$$

$$= P\{x^2+y^2 \leq z^2\}$$

$$= \iint_G \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}} dx dy$$

$$= \int_0^{2\pi} d\theta \int_0^z \frac{1}{2\pi} e^{-\frac{r^2}{2}} \cdot r dr$$

$$= 1 - e^{-\frac{z^2}{2}}$$

分布: $F_Z(z) = \begin{cases} 1 - e^{-\frac{z^2}{2}} & z \geq 0 \\ 0 & z < 0 \end{cases}$

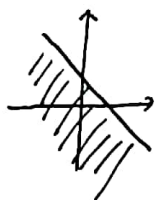
↓ 求导

密度 $f_Z(z) = \begin{cases} ze^{-\frac{z^2}{2}} & z \geq 0 \\ 0 & z < 0 \end{cases}$

例: (1)

$$Z = X + Y, \text{ 求 } F_Z(z)$$

$$F_Z(z) = P\{Z \leq z\} = P\{X + Y \leq z\}$$



$$= \iint_{x+y \leq z} f(x, y) dx dy$$

$$= \int_{-\infty}^{+\infty} dx \int_{-\infty}^{z-x} f(x, y) dy$$

设 $t = x + y$.

$$(X \text{ 型}) = \int_{-\infty}^{+\infty} dx \int_{-\infty}^z f(x, t-x) d(t-x)$$

$$(Y \text{ 型}) = \int_{-\infty}^z \left[\int_{-\infty}^{+\infty} f(x, t-x) dx \right] dt$$

$$f_Z(z) = \int_{-\infty}^{+\infty} f(x, z-x) dx$$

$$f_Z(z) = \int_{-\infty}^{+\infty} f(z-y, y) dy$$

若独立, 则联合密度可写为边缘密度.

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x) dx \quad \text{卷积公式}$$

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(z-y) f_Y(y) dy$$

条件 ① $Z = X + Y$. ② X, Y 独立

例 (2) $X \sim N(0, 1) \quad Y \sim N(0, 1)$

X, Y 独立, $Z = X + Y$.

解: $\phi_Z(z) = \int_{-\infty}^{+\infty} \phi_X(x) \phi_Y(z-x) dx$

$$= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \frac{1}{\sqrt{2\pi}} e^{-\frac{(z-x)^2}{2}} dx$$

$$= \int_{-\infty}^{+\infty} \frac{1}{2\pi} e^{-\frac{z^2}{4}} e^{-(x-\frac{z}{2})^2} dx$$

$$= \frac{1}{2\pi} e^{-\frac{z^2}{4}} \int_{-\infty}^{+\infty} e^{-(x-\frac{z}{2})^2} d(x-\frac{z}{2})$$

$$= \frac{1}{\sqrt{2\pi} \cdot \sqrt{2}} e^{-\frac{z^2}{4}}$$

$\therefore Z \sim N(0, 2)$ 结论(直接记取)



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⑧

$$X \sim N(\mu_1, \sigma_1^2) \quad Y \sim N(\mu_2, \sigma_2^2)$$

$$X+Y \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$$

例 $X \sim N(2, 5) \quad Y \sim N(3, 9)$

$$X+Y \sim N(5, 14)$$

$$M = \max\{X, Y\} \quad N = \min\{X, Y\}$$

$$\{\max\{X, Y\} \leq z\} = \{X \leq z, Y \leq z\}$$

$$F_M(z) = P\{M \leq z\} = P\{X \leq z, Y \leq z\}$$

$$= P\{X \leq z\} P\{Y \leq z\}$$

$$= F_X(z) F_Y(z)$$

$$F_N(z) = P\{N \leq z\} = 1 - P\{N > z\}$$

$$= 1 - P\{X > z, Y > z\}$$

$$= 1 - P\{X > z\} P\{Y > z\}$$

$$= 1 - [1 - P\{X \leq z\}][1 - P\{Y \leq z\}]$$

$$= 1 - [1 - F_X(z)][1 - F_Y(z)]$$

例: X, Y 独立. $X: [0, 1]$ 均匀分布

$Y: \lambda=3$ 的指数分布. 求

$$M = \max(X, Y) \quad N = \min(X, Y)$$

解: $f_X(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{else} \end{cases}$

$$f_Y(y) = \begin{cases} 3e^{-3y} & y > 0 \\ 0 & y \leq 0 \end{cases}$$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases}$$

$$F_Y(y) = \begin{cases} 1 - e^{-3y} & y > 0 \\ 0 & y \leq 0 \end{cases}$$

$$M = \max\{X, Y\}$$

$$F_M(z) = F_X(z) F_Y(z)$$

求导 $\downarrow$

$$= \begin{cases} 0 & z < 0 \\ 2(1-e^{-3z}) & 0 \leq z < 1 \\ 1-e^{-3z} & z \geq 1 \end{cases}$$

$$f_M(z) = \begin{cases} 0 & z < 0 \\ 1-e^{-3z} + 3ze^{-3z} & 0 \leq z < 1 \\ 3e^{-3z} & z \geq 1 \end{cases}$$

$$N = \min\{X, Y\}$$

$$F_N(z) = \begin{cases} 0 & z < 0 \\ \frac{1-(1-z)e^{-3z}}{1} & 0 \leq z < 1 \\ 1 & z \geq 1 \end{cases}$$

$$f_N(z) = \begin{cases} 4e^{-3z} - 3ze^{-3z} & 0 \leq z < 1 \\ 0 & \text{else} \end{cases}$$



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